

Convert EMCAD and POLYPPVT Model into ONNX

EMCAD: we convert it successfully without changing any function.

Test the Model accuracy change(ONNX vs torch)
in python with GPU RTX A100

EMCAD Difference

Convert ONNX to TensorRT aarch64 chipset.

Jetpack 5.1.3

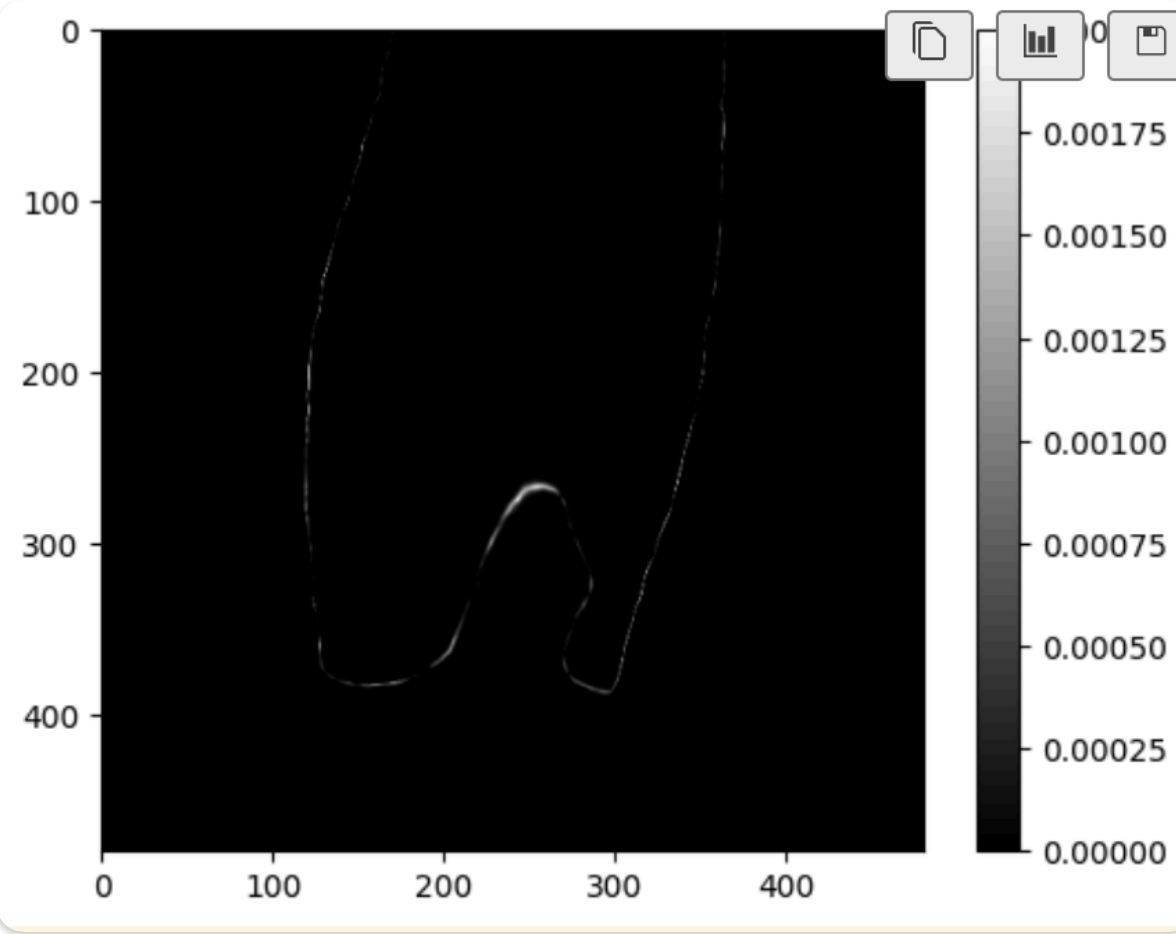
POLYPPVT: We change three line in pvt.py

```
self.avg_pool = nn.AdaptiveAvgPool2d(1)
```

```
self.avg_pool = nn.AvgPool2d(kernel_size=120, stride=120, padding=0)
```

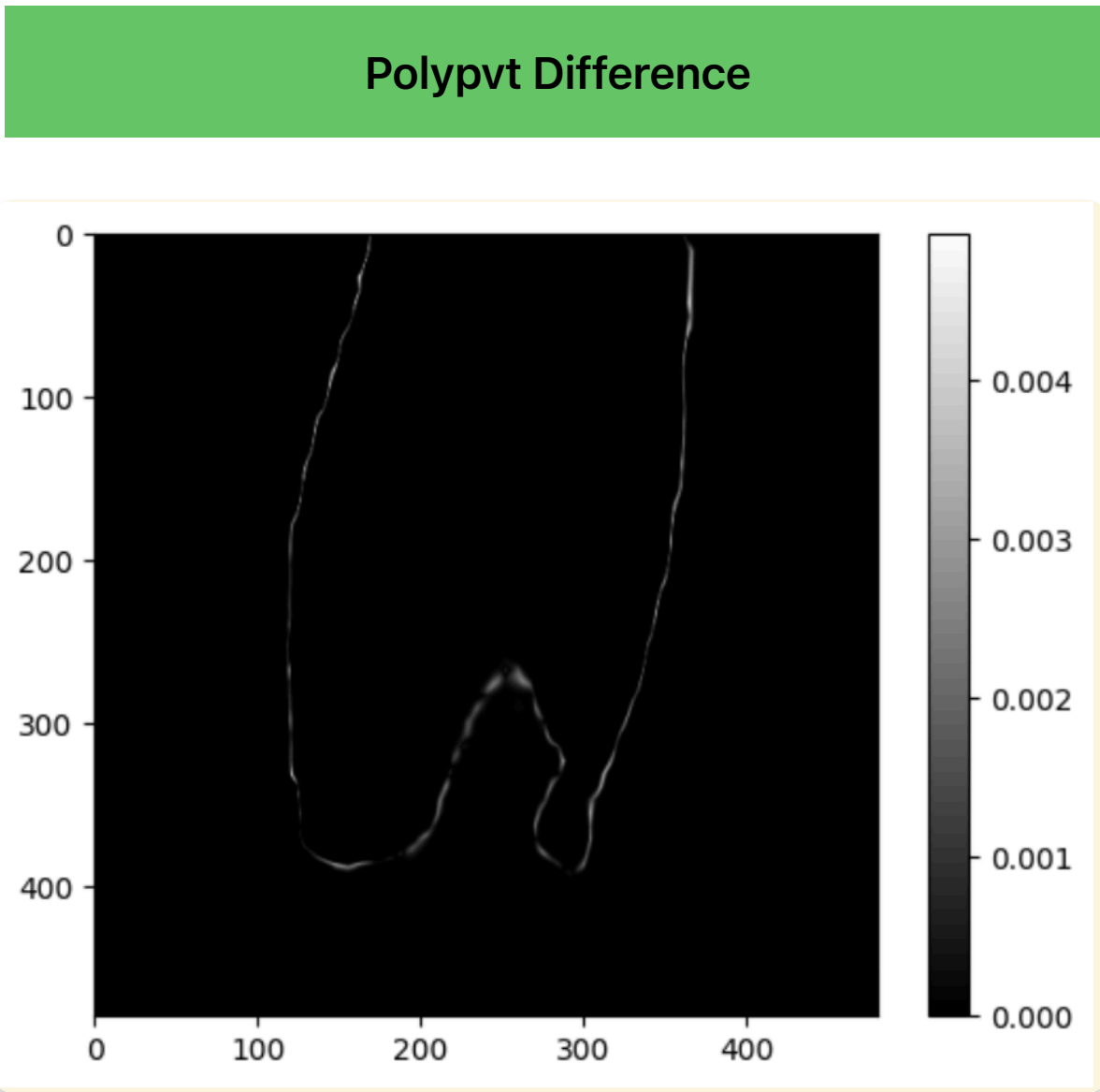
```
self.max_pool = nn.AdaptiveMaxPool2d(1)
```

```
self.max_pool = nn.MaxPool2d(kernel_size=120, stride=120, padding=0)
```



```
self.priors = nn.AdaptiveAvgPool2d(output_size=(mids + 2, mids + 2))
```

```
self.priors = nn.AvgPool2d(kernel_size=10, stride= 10, padding=0)
```



05/25/2025: Ziping help to update the model definition, we run second round ONNX model convert.

Install softmax plugin for TensorRT, deserializeCudaEngine

```
TensorRT 8.5: LayerNormalization support issue.
op_type: "LayerNormalization"
  attribute {
    name: "axis"
    i: -1
    type: INT
  }
  attribute {
    name: "epsilon"
    f: 1e-05
    type: FLOAT
  }
}
```

[05/20/2025-18:48:35] [E] [TRT] ModelImporter.cpp:729: --- End node ---
[05/20/2025-18:48:35] [E] [TRT] ModelImporter.cpp:731: ERROR: builtin_op_importers.cpp:5427 In function importFallbackPluginImporter:
[05/20/2025-18:48:35] [E] Failed to parse onnx file
[05/20/2025-18:48:35] [I] Finish parsing network model
[05/20/2025-18:48:35] [E] Parsing model failed
[05/20/2025-18:48:35] [E] Failed to create engine from model or file.
[05/20/2025-18:48:35] [E] Engine set up failed
&&& FAILED TensorRT.trtexec [TensorRT v8502] # /usr/src/tensorrt/bin/trtexec --onnx=onnx/emcad_0.onnx --saveEngine=tensorrt/emcad_0.trt

Request to change the model:
1. Combine model output
2. Move nn.softmax step into the model definition.

Jetpack 6.2

Good for dummy run

Build the TensorRT inferencing Tool on Jetpack

1. Build Phase:
Create a network definition.
Specify a configuration for the builder.
Call the builder to create the engine.

Build EMCAD 3 TensorRT Engine(6.6GB + 1.3GB = 7.8GB)

```
DLDesignerWorker export-rt --use-system-rt --config /tmp/NVIDIA_Night Deep Learning Designer-cklN/rtconfig.json --save-rt-data /home/liu/Desktop/Burn_model/ONNX/updated_model/emcad3.json /home/liu/Desktop/Burn_model/ONNX/updated_model/emcad_3.onnx /home/liu/Desktop/Burn_model/ONNX/updated_model/emcad3.trt
```

Launch succeeded.
[INF] [TRT] [MemUsageChange] Init CUDA: CPU +2, GPU +0, now: CPU 354, GPU 5881 (MiB)
[INF] [TRT] [MemUsageChange] Init builder kernel library: CPU +927, GPU +784, now: CPU 1324, GPU 6630 (MiB)
[INF] [TRT] [TRT] Input filename: /home/liu/Desktop/Burn_model/ONNX/updated_model/emcad_3.onnx
[INF] [TRT] ONNX IR version: 0.0.8

Build EMCAD 4 TensorRT Engine(7.5GB + 1.3GB = 8.8GB)

```
Using cuda-rt as engine and system-rt as plugin.  
Launched process: DLDesignerWorker (pid: 22364)
```

DLDesignerWorker export-rt --use-system-rt --config /tmp/NVIDIA_Night Deep Learning Designer-cklN/rtconfig.json --save-rt-data /home/liu/Desktop/Burn_model/ONNX/updated_model/emcad4.json /home/liu/Desktop/Burn_model/ONNX/updated_model/emcad_4.onnx /home/liu/Desktop/Burn_model/ONNX/updated_model/emcad4.trt

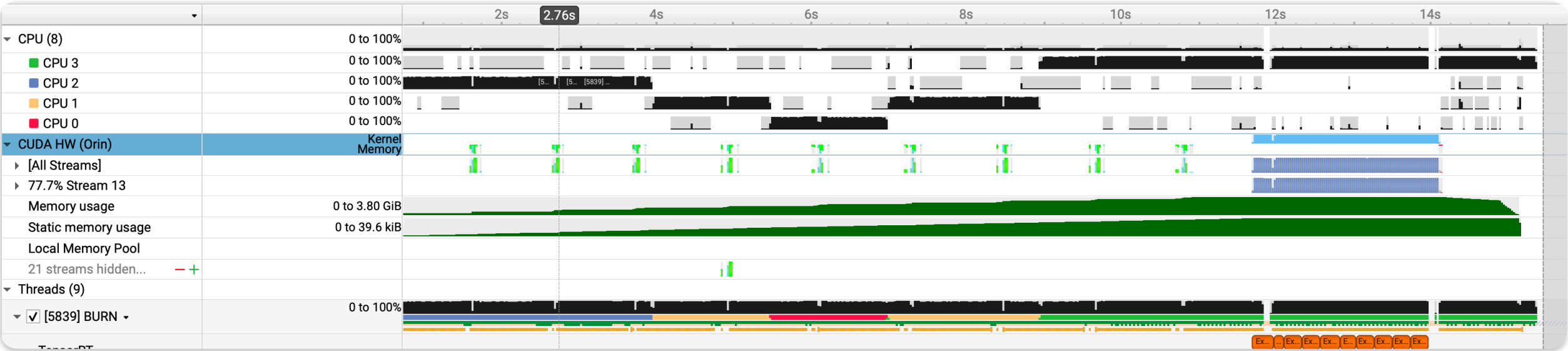
Launch succeeded.
[INF] [TRT] [MemUsageChange] Init CUDA: CPU +2, GPU +0, now: CPU 355, GPU 6770 (MiB)
[INF] [TRT] [MemUsageChange] Init builder kernel library: CPU +927, GPU +736, now: CPU 1325, GPU 7551 (MiB)

Build PVT TensorRT Engine use similar amount of memory.

TODO: DLA, will do it if I have time

Inference Face:
1. Create runtime to store weight and layer
2. Create CudaEngine pointer.
3. Create Execution Context for CUDA stream inference.

```
INFO: [MemUsageChange] TensorRT-managed allocation in IExecutionContext creation: CPU +0, GPU +70, now: CPU 10, GPU 3492 (MiB)  
VERBOSE: CUDA lazy loading is enabled.  
[INF] Engine 9 loaded successfully.  
[INF] Loading Engine 10 from: /home/liu/Desktop/TORCHCPP_BURN/engine/pvt4.trt  
INFO: Loaded engine size: 329 MiB  
WARNING: Using an engine plan file across different models of devices is not recommended and is likely to affect performance or even cause  
VERBOSE: Deserialization required 182626 microseconds.  
VERBOSE: Total per-runner device persistent memory is 0  
VERBOSE: Total per-runner host persistent memory is 652432  
VERBOSE: Allocated device scratch memory of size 73339904  
VERBOSE: - Runner scratch: 73339904 bytes  
INFO: [MemUsageChange] TensorRT-managed allocation in IExecutionContext creation: CPU +0, GPU +70, now: CPU 11, GPU 3873 (MiB)  
VERBOSE: CUDA lazy loading is enabled.  
[INF] Engine 10 loaded successfully.  
Time taken to load engine: 11335 ms  
Running prediction...
```



OPENCV resize

Use torch resize method

